

TASK1: Data Cleaning and Preprocessing

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Task breakdown:

- Loaded the given dataset
- Found 11 hidden missing values and fixed in TotalCharges column
- Converted Churn column from text(YES/NO) to numerical values(1/0)
- Implemented one-hot encoding to 8 categorical columns
- Saved the cleaned data as 'Cleaned_Churn_Data.csv'

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Matplotlib is building the font cache; this may take a moment.

```
In [4]: import os
print(os.getcwd())
```

C:\Users\thend\OneDrive\Desktop\Telco Customer Churn

```
In [6]: df = pd.read_csv('Telco_Customer_Churn_Dataset.csv')
df.head()
```

Out[6]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mul
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns



```
In [7]: print("Rows and Columns:", df.shape)
print("\nColumn names:")
print(df.columns.tolist())
```

Rows and Columns: (7043, 21)

Column names:

['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn']

```
In [8]: print("Missing values per column:")
print(df.isnull().sum())
```

Missing values per column:

```
customerID      0
gender          0
SeniorCitizen   0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
MonthlyCharges  0
TotalCharges    0
Churn           0
dtype: int64
```

```
In [9]: print("Unique error values in TotalChargers Column:")
print(df[df['TotalCharges'] == ' ']['TotalCharges'].count())
```

Unique error values in TotalChargers Column:
11

```
In [10]: print(df[df['TotalCharges'] == ' '][['customerID', 'tenure', 'MonthlyCharges', 'TotalCharges']])
```

	customerID	tenure	MonthlyCharges	TotalCharges
488	4472-LVYGI	0	52.55	
753	3115-CZMZD	0	20.25	
936	5709-LVOEQ	0	80.85	
1082	4367-NUYAO	0	25.75	
1340	1371-DWPAZ	0	56.05	
3331	7644-OMVMY	0	19.85	
3826	3213-VVOLG	0	25.35	
4380	2520-SGTTA	0	20.00	
5218	2923-ARZLG	0	19.70	
6670	4075-WKNIU	0	73.35	
6754	2775-SEFEE	0	61.90	

```
In [13]: df['TotalCharges'] = df['TotalCharges'].replace(' ', '0')
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'])
print("TotalCharges error has been fixed! Here is what's been fixed:")
print(df[df['tenure'] == 0][['customerID', 'tenure', 'MonthlyCharges', 'TotalCharges']])
```

TotalCharges error has been fixed! Here is what's been fixed:

	customerID	tenure	MonthlyCharges	TotalCharges
488	4472-LVYGI	0	52.55	0.0
753	3115-CZMZD	0	20.25	0.0
936	5709-LVOEQ	0	80.85	0.0
1082	4367-NUYAO	0	25.75	0.0
1340	1371-DWPAZ	0	56.05	0.0
3331	7644-OMVMY	0	19.85	0.0
3826	3213-VVOLG	0	25.35	0.0
4380	2520-SGTTA	0	20.00	0.0
5218	2923-ARZLG	0	19.70	0.0
6670	4075-WKNUI	0	73.35	0.0
6754	2775-SEFEE	0	61.90	0.0

```
In [15]: df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})
print("Churn column converted:")
print(df['Churn'].value_counts())
print("\nSample:")
print(df['Churn'].head(10))
```

Churn column converted:

Churn

0 5174

1 1869

Name: count, dtype: int64

Sample:

0 0

1 0

2 1

3 0

4 1

5 1

6 0

7 0

8 1

9 0

Name: Churn, dtype: int64

```
In [16]: df_encoded = pd.get_dummies(df, columns=['Contract', 'PaymentMethod',
                                                'InternetService', 'gender',
                                                'Partner', 'Dependents',
                                                'PhoneService', 'PaperlessBilling'],
                                           drop_first=True)
print("Original columns: 21")
print("New shape after encoding:", df_encoded.shape)
print("\nNew column names:")
print(df_encoded.columns.tolist())
```

Original columns: 21

New shape after encoding: (7043, 25)

New column names:

['customerID', 'SeniorCitizen', 'tenure', 'MultipleLines', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'MonthlyCharges', 'TotalCharges', 'Churn', 'Contract_One year', 'Contract_Two year', 'PaymentMethod_Credit card (automatic)', 'PaymentMethod_Electronic check', 'PaymentMethod_Mailed check', 'InternetService_Fiber optic', 'InternetService_No', 'gender_Male', 'Partner_Yes', 'Dependents_Yes', 'PhoneService_Yes', 'PaperlessBilling_Yes']

```
In [17]: df_encoded.to_csv('Cleaned_Churn_Data.csv', index=False)

print("Cleaned data saved as: 'Cleaned_Churn_Data.csv'")
print("Task 1 Complete:")
print(f"\nFinal dataset: {df_encoded.shape[0]} customers, {df_encoded.shape[1]}")
```

Cleaned data saved as: 'Cleaned_Churn_Data.csv'

Task 1 Complete:

Final dataset: 7043 customers, 25 columns